
SciMiner

-Server version installation guide-

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DISCLAIMER

The use of SciMiner is strictly limited to academic research purposes. SciMiner must not be used for any commercial purpose. We, the developers of the SciMiner, and the University of Michigan, Ann Arbor are not responsible in away for any trouble that might be caused by inappropriate use of the SciMiner system. It must be noted that the copyrights of the processed articles belong to their respective publishers not to the SciMiner users or developers.

SYSTEM REQUIREMENT

For smooth running of SciMiner at an institution level, it is recommended that SciMiner be installed on a workstation level machine with UNIX (or Linux) operating system with at least 2 GB of RAM. Multi-core or Multi-CPU machines with 4 GB of RAM are preferred. Hard disk drive can be quickly filled up with multiple uses of big corpus. 10GB of free space is the absolute minimum space but at least 100GB is recommended for 500,000 documents.

[Fedora](#) 6, 7, 8 and 9 have been extensively used for SciMiner. Thus, we do not expect any issue in installing SciMiner on these platforms. Other Linux platform should also work fine. For windows platform, [VMware](#) virtual system is recommended. The SciMiner Package II is provided as a virtual disk, which can be readily used by [VMware Player](#).

If you choose to install the Package I on any system, required CPAN Perl modules, MySQL, and a web-server should be working properly before using SciMiner.

The public version of SciMiner is running on a machine with two Intel Xeon CPUs, 3GB RAM, and 900 GB of Hard disk drive with [Fedora](#) 7.

TYPES OF DISTRIBUTION PACKAGES

Two different types of distribution packages are available for download. Experience in using Linux is strongly required. If you have little or no experience in using Linux, or just want to test SciMiner on your personal computer, we strongly recommend that you download the **Package II**.

Package I: Basic package

This comes in a tar.gz file, which contains all of the necessary scripts and modules. But required CPAN Perl modules, MySQL, and a web-server (lighttpd or any) should be taken care of separately. You have to create ‘**sciminer**’ account and uncompress this package on the base directory of ‘**sciminer**’.

Package II: Fedora 8 Linux virtual ready-to-use image

This comes in a form of VMware virtual disk which has Fedora 8 with SciMiner installed. This can be used with either VMware Workstation or VMware Player (Free of charge). Both can be obtained from <http://www.vmware.com/>

This virtual disk does NOT come with a pre-allocated disk space. Thus the actual disk size grows as the Fedora is used. This virtual disk comes with ‘Split disk into 2 GB files’.

For a better performance, it is recommended that the virtual disk is converted ‘**pre-allocated disk space**’ mode and increase the overall hard disk size if you would like to use SciMiner seriously. You can refer to the following web document.

http://www.vmweekly.com/ask_an_expert/how-can-i-expand-disk-size-and-preallocate-all-the-space-working-with-vmware-workstation-6/1/

Default account name and password

‘root’ account password:	124356
‘ sciminer ’ account password:	124356
MySQL ‘ sciminer ’ account password:	124356

HOW TO USE THE PACKAGE II

The Package II comes in two compressed files.

STEP1) Once downloads are completed, unzip them in a same directory. You might need to install compression utilities like [WinZip](#) or [WinRar](#) which support zip compression format.

STEP2) If your computer does not have **VMware workstation** or **VMplayer** installed, go to the [VMware web site](#) and download either version. Note that **VMplayer** is free of charge.

STEP3) Use ‘Open’ command and navigate to the directory where the SciMiner disks have been unzipped. Select ‘Other Linux 2.6.x kernel.vmx’ file.

STEP4) Login as ‘sciminer’

STEP5) Launch a terminal (Applications > Systems Tools > Terminal) and type ‘startweb’ to initiate the local SciMiner web server.

STEP6) Launch a web-browser and go to “http://localhost:8888/SciMiner/”

REQUIRED PROGRAMS AND PERL MODULES

There are some applications and Perl modules required for SciMiner to run. If you have downloaded the **Package I**, please make sure that all of the following items are available and correctly configured.

I Required Applications

The following programs need to be available on the system before using SciMiner. Their absolute path and login information (MySQL) should be correctly specified in `/home/sciminer/ANNOTATION/SciMinerDB/annotationENV.ini`

n [MySQL database](#)

u Account: sciminer

u Password: 124356

n [ImageMagick](#)

n **Web-server (eg. Lighttpd)**

u CGI must be enabled.

u Port: Default 8888

n **Perl**

I Required Perl Modules

The following Perl modules are needed on the system before using SciMiner. You can download and install them from CPAN (<http://www.cpan.org/>) or use 'cpan' from the Linux command line. If 'cpan' is not available on your system, you can install it by `>yum install perl-CPAN` (on Fedora Linux systems). Please, refer to <http://perl.about.com/od/packagesmodules/qt/perlcpn.htm> or Google 'Installing Perl module from CPAN' for how to use cpan shell. This is to be done as **root** account.

n Boulder::Medline

u You need to modify this module at Line 274: Modify
@recorlines[\$i] to \$recordlines[\$i]

n Text::NSP

n CGI::Debug

n CGI::Simple

n CGI::Session

n CGI::Application

n HTML::Template

n Data::Dumper

n Unicode::String

n XML::XPath

POST-INSTALLATION PROCEDURES

After you successfully installed or ran the VMWare Fedora 8 virtual machine with SciMiner, you now have to modify the main configuration file and web htmls according to your or your institutional information.

STEP1) Change the **passwords** for the **root** and **sciminer** account (only for virtual machine)

>passwd will guide you to change the passwords.

STEP2) Modify the **main configuration file**

‘/home/sciminer/ANNOTATION/SciMinerDB/annotationENV.ini’. Items in bold are strongly recommended to be changed.

```
[SciMinerDB]
OS=linux
ANNOPath=/home/SciMiner/ANNOTATION/
SciMinerPath=/home/SciMiner/ANNOTATION/SciMinerDB/
SciMinerWebPath=/home/SciMiner/web/html/SciMiner/
SciMinerWebTempPath=/tmp/SciMiner/
Perl=/usr/bin/perl
Institution=University of Michigan, Ann Arbor
DB=sciminer
username=sciminer
password=124356
SciMinerServerURL=http://localhost:8888/
WholePMID=17760657
```

STEP3) Modify the HTML files at /home/sciminer/web/html/SciMiner/

- 1) You must put your **email** at the line 21 of the following files. Modify mailto:? to mailto:<<YOUR EMAIL HERE>>?.
 - A. analysisIndex.tmpl
 - B. completedIndex.tmpl
 - C. SciMinerIndex.tmpl
- 2) Other web pages like main2.html, intro2.html, and contact2.html can also be freely changed as you need.

STEP4) Modify the **\$adminEmail** to your email address at line 25 of **'~/ANNOTATION/SciMinerDB/Modules/Annotation/SciMiner.pm'** in order for you to get an email regarding any system issue. You

```
my $adminEmail = '<<YOUR EMAIL HERE>>';
```

STEP5) Almost every Perl scripts, modules, and CGI scripts used in the SciMiner system has the following environment statement at the beginning of each file.

```
BEGIN {push (@INC, "/home/sciminer/ANNOTATION/SciMinerDB/Modules/");}
```

This is to specify the location of the required SciMiner modules. If you installed SciMiner on different account or location, you must modify this line according to your current system configuration.

ADDITIONAL STEPS FOR PACKAGE I

STEP6) If you downloaded the Package I, it should be unpacked onto **/home/sciminer/** . Create 'SciMiner' directory at the temporary directory and change the permission to 777

```
>mkdir /tmp/SciMiner/  
>chmod 777 SciMiner
```

STEP7) The following directories should have 777 permission since they need to be accessible by web server and results files are created.

```
>chmod 777 -R ~/ANNOTATION/SciMiner/Work/  
>chmod 777 ~/web/html/SciMiner/FinalResults/
```

ADDITIONAL NOTES

-- EMAIL NOTICE --

SciMiner tries to send out an email notice when queued jobs are completed. In order to use this email notice, however, you must have your machine configured correctly for outgoing emails. Every system has different configuration depending on their email servers. Thus you have to get help the system administrators at your institution. It should be noted that the **Package II** does **NOT** come with preconfigured outgoing email service enabled.

If you plan to deploy SciMiner as an accessible system from other users in your institution, you have to get a static IP and configure the sendmail correctly. We strongly recommend you to seek help from the system administrators at your institution.

Even if email notice is not configured on your SciMiner, users can always check all of their completed jobs from the 'Completed' menu.

-- POP-UP BLOCKING --

Pop-up blocking option must be disabled from the client side. Most of the recent web-browsers these days have pop-up disabled. If you click a link on SciMiner result pages and a new window or tab pops up then closes right away, it is very likely that your browser is blocking pop-up on your server.

HOW TO START SCIMINER SERVER

Once you have completed necessary changes above, you are now ready to start the SciMiner server.

If a web server (lighttpd, Apache, or any) is already running, launch a web browser and try respective URL.

If you are using the configuration as provided, you can simply start the web server by the following command.

>startweb

This command is only applicable to the SciMiner system and is comparable to
'/usr/sbin/lighttpd -f /home/SciMiner/web/lighttpd/lighttpd.conf'

HOW TO CONFIRM SCIMINER IS RUNNING CORRECTLY

Open a web browser and go to <http://localhost:8888/SciMiner/> if it is in the Fedora8 virtual machine. If you are connecting from outside the virtual machine, you have to specify the correct URL (probably with IP address) to the installed SciMiner. If the web server is running, you will be able to see the main page of the SciMiner server.

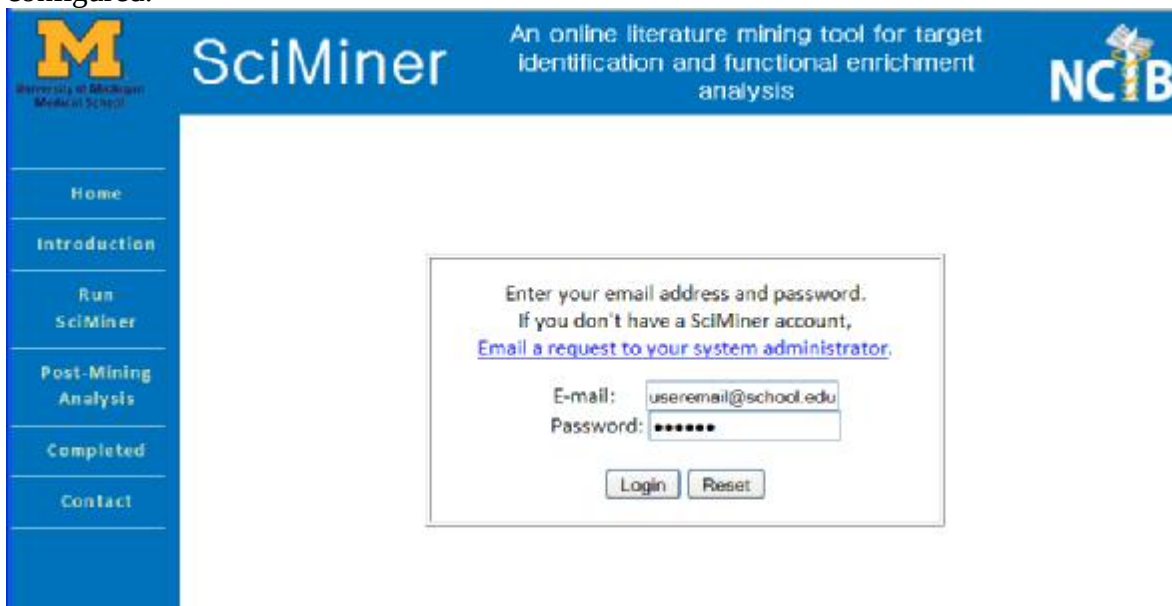


Currently, the network of the Fedora 8 virtual machine with SciMiner is configured with **NAT**, meaning that the virtual machine shares your host computer's network. Since your host computer is working as a DHCP server and the Fedora 8 is on the private network under the DHCP server, users from the outside the host computer **cannot** directly access the Fedora 8.

There could several possible solutions to this problem if you want the SciMiner accessible from outside.

- 1) Set the VMware network use '**Bridged**' network instead of '**NAT**'. It makes the virtual machine get an IP from the outside DHCP server, which your host machine is physically connected.
- 2) Get a static IP from your institution's network administrator and use that IP information. You may have to select '**Custom**' instead of the '**Bridged**' or '**NAT**'. And set the IP information within the Fedora.
Menu > Administration > Network
- 3) Install the **Package I** on a machine that can be accessed from outside.

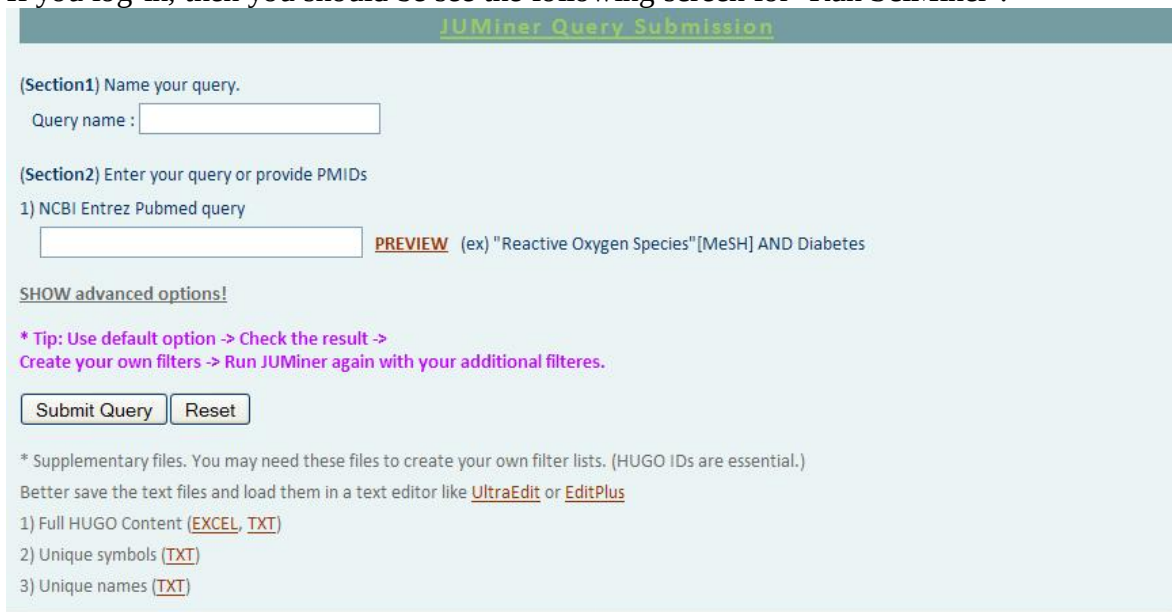
If you click, 'Run SciMiner', 'Post-Mining Analysis', or 'Completed', you should be able to see the following login screen. If not, some SciMiner components are not correctly configured.



The image shows the SciMiner login interface. At the top, there is a blue header with the SciMiner logo (a large 'M' with 'University of Michigan Medical School' below it) on the left, the text 'SciMiner' in the center, and 'An online literature mining tool for target identification and functional enrichment analysis' on the right. The NCBI logo is also visible on the far right. A vertical blue sidebar on the left contains links: Home, Introduction, Run SciMiner, Post-Mining Analysis, Completed, and Contact. The main content area is white and contains a login form with the text: 'Enter your email address and password. If you don't have a SciMiner account, Email a request to your system administrator.' Below this text are input fields for 'E-mail:' (containing 'useremail@school.edu') and 'Password:' (containing six dots). At the bottom of the form are 'Login' and 'Reset' buttons.

Enter your registered email address and password to log-in. Users should send email to the system administrator (who installs the SciMiner) to request an account on SciMiner.

If you log-in, then you should be see the following screen for 'Run SciMiner'.



The image shows the JUMiner Query Submission interface. At the top, there is a green header with the text 'JUMiner Query Submission'. Below this, the interface is divided into sections. Section 1 is titled '(Section1) Name your query.' and contains a text input field for 'Query name :'. Section 2 is titled '(Section2) Enter your query or provide PMIDs' and contains a text input field for '1) NCBI Entrez Pubmed query'. Below this input field is a 'PREVIEW' button and the text '(ex) "Reactive Oxygen Species"[MeSH] AND Diabetes'. Below Section 2 is a link 'SHOW advanced options!'. Below this link is a tip: '* Tip: Use default option -> Check the result -> Create your own filters -> Run JUMiner again with your additional filteres.' Below the tip are 'Submit Query' and 'Reset' buttons. At the bottom, there is a note: '* Supplementary files. You may need these files to create your own filter lists. (HUGO IDs are essential.) Better save the text files and load them in a text editor like UltraEdit or EditPlus'. Below this note are three links: '1) Full HUGO Content (EXCEL, TXT)', '2) Unique symbols (TXT)', and '3) Unique names (TXT)'.

HOW TO ADD A NEW USER

New users can be added to the SciMiner system by inserting a new record to 'user' table in the 'SciMiner' db of the MySQL database. Just for your reference, here is the simplest way of adding one user.

- 1) First log-in to MySQL
 `>mysql -usciminer -p124356 sciminer`
- 2) Insert a new user information
 `mysql> insert into user
 (email, name, passCode, maxDoc, maxNewDoc, editLevel, institute,
 deptOrLab) values ("user@email.address", "Mr. SciMiner", "new
 password", 3000, 1000, 3, "User institution", "User Lab");`

PremiumSoft's Navicat for MySQL would be very useful and easy to use for managing users. It can be downloaded at <http://mysql.navicat.com/index.html>.

<Fields>

maxDoc	: The number of maximum documents per single query
maxNewDoc	: The number of maximum NEW documents per single query
editLevel	: The current version of SciMiner is not actively using this field. In the future version, editLevel will specify different levels of edit function for each user.
institute	: User's institution
deptOrLab	: User's lab or department

3000 for maxDoc and 1000 for maxNewDoc would be recommended. Generally, a maximum of 20,000 is acceptable for the maxDoc on a machine with 2GB RAM. System administrator can increase or decrease this number depending on your system's capacity.

CHECKING LOG FILES

The usage of SciMiner is logged onto the following three files at the “/tmp/” directory.

/tmp/SciMinerQuery.log	: For SciMiner Mining Query
/tmp/SciMinerGRIFQuery.log	: For Gene2PubMed / GeneRIF Query
/tmp/SciMinerAnalysis.log	: For Post-Mining Analysis Query

Aliases are set to give a quick access to the log files.

- ‘**tm**’ for SciMinerQuery.log
- ‘**tg**’ for SciMinerGRIFQuery
- ‘**ta**’ for SciMinerAnalysis.log

Each will show the last 300 lines of the log.

CONTACTS

If you encounter any problem in deploying the server version of SciMiner or have any questions or comments, please contact Junguk Hur.

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